
AMORTIZED INFERENCE FOR STOCHASTIC BIOLOGICAL MODELS USING NEURAL POSTERIOR ESTIMATION

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ABSTRACT

Parameter inference for stochastic models is essential for quantitative biology but is hampered by intractable likelihoods. Agent-based random walk models, used widely in ecology and cell biology, exemplify this challenge, often forcing researchers to use surrogate models or likelihood-free methods like Approximate Bayesian Computation (ABC) that introduce biases and tuning difficulties. Here, we overcome these limitations using Neural Posterior Estimation (NPE), a simulation-based, machine learning inference framework that learns the full posterior distribution. Once trained, NPE provides amortized, near-instantaneous posterior estimates. We demonstrate this framework in two settings: first, using one-dimensional summary statistics (column counts), and second, via a novel convolutional neural network (CNN) that learns directly from raw two-dimensional spatial data. On synthetic barrier assay data, our NPE pipeline matches or exceeds the precision of established baselines at a substantially reduced computational cost per inference. The spatial CNN approach further reduces posterior uncertainty by 8-9%, highlighting a critical trade-off between statistical precision and the increased computational cost. Our analysis provides a clear framework for deciding when the full spatial data is necessary versus when summary statistics suffice, a finding with direct implications for experimental design. We provide an open-source implementation of our pipeline to facilitate its adoption and extension to more complex, spatially-structured biological models.

Keywords First keyword · Second keyword · More

1 Introduction

Parameter inference is the critical link between mathematical models and biological understanding, enabling quantitative insights from noisy experimental data. Inference methods provide researchers with a powerful toolkit to generate predictions about system behavior, estimate unknown parameters, test competing hypotheses, and rigorously quantify uncertainty. These applications are critical across a wide range of fields, including epidemiology [Dunson, 2005, Semochkina and Walsh, 2025], ecology [Qian et al., 2009, Rosen et al., 2001], cellular biology [Vallejos et al., 2015], gene regulatory networks [Liu et al., 2016, Wang et al., 2019], systems biology [Rabinowitz and Sridhar, 2013] phylogenetics [Nascimento et al., 2017], and neuroscience [Coventry and Bartlett, 2024].

For deterministic continuum models described by ordinary and partial differential equations, inference methodologies are mature and well-established. Both Bayesian [Bernardo and Smith, 1994, Gelman et al., 1995] and frequentist [Cox and Hinkley, 1974, Casella and Berger, 1990] approaches provide robust foundations for parameter estimation, model validation, and uncertainty quantification. The sophistication of these inference techniques has enabled reliable parameter estimation and prediction across numerous scientific disciplines including climate science [Vrugt et al., 2014], cosmology [Planck Collaboration et al., 2020], fluid dynamics [Cheung et al., 2011] gravitational wave astrophysics [Veitch et al., 2015].

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Stochastic biological models, however, present fundamental inferential challenges that resist these established approaches. Agent-based models and discrete stochastic processes, while capturing essential biological variability and heterogeneity, typically possess intractable likelihoods that preclude standard statistical methods. Experimental protocols exhibit substantial variation, datasets are frequently sparse and noisy, and the stochastic nature of biological processes introduces complex dependencies that deterministic frameworks cannot adequately represent. These characteristics create a disconnect between the sophisticated stochastic models needed to represent biological reality and the inference tools available to calibrate them from data.

Several recent developments have attempted to address these inferential challenges while preserving the biological realism of stochastic models [Carr et al., 2021, Simpson et al., 2022, Simpson and Plank, 2025], typically focusing on inference, identifiability, and prediction for stochastic agent-based models of migration and proliferation. These approaches often use experimental data such as scratch (barrier) assay experiments to calibrate high-fidelity discrete random-walk models, highlighting applications where stochasticity is visible and decision-relevant. At the same time, these successes have clarified several canonical challenges that have historically limited the broader application of stochastic models. While these approaches demonstrate the potential of sophisticated inference frameworks, they also reveal fundamental limitations that must be addressed to fully realize the promise of stochastic biological modeling. These canonical challenges can be summarized as follows.

1. **Intractable likelihoods:** The likelihood function for most agent-based models is intractable, precluding the use of standard methods like Maximum Likelihood Estimation or many Markov chain Monte Carlo (MCMC) algorithms.
2. **Surrogate-model error:** A common strategy to make inference tractable is to replace the computationally expensive stochastic simulator with a cheap surrogate, typically a deterministic Partial Differential Equation (PDE) describing mean-field behavior. While this enables efficient MCMC or MLE techniques, the resulting inference is conditional on the surrogate being a faithful approximation, an assumption that, if violated, can lead to biased parameter estimates and inaccurate predictions.
3. **Noise-model misspecification:** The use of a deterministic surrogate creates the problem of specifying a noise model to connect its clean output to noisy data. A misspecified choice can lead to non-physical predictions and poorly calibrated uncertainties.
4. **Likelihood-free inefficiency:** Classical likelihood-free alternatives like Approximate Bayesian Computation (ABC) are notoriously computationally expensive and sensitive to the choice of summary statistics, distance metrics, and tolerance thresholds.
5. **Information loss from summary statistics:** Classical methods often require reducing complex, high-dimensional data (e.g., the full 2D spatial arrangement of cells) to lower-dimensional summary statistics (e.g., a 1D cell density profile). This step risks a significant loss of information, and the choice of statistics can heavily bias the inference.

In this paper, we propose that these fundamental challenges can be addressed in a unified manner using Neural Posterior Estimation (NPE), a modern simulation-based inference (SBI) paradigm. NPE leverages flexible neural density estimators to learn the posterior distribution $p(\theta | x)$ directly from simulator-generated (θ, x) pairs. This approach offers direct remedies to each of the aforementioned challenges.

1. It is likelihood-free, circumventing intractability.
2. It learns directly from the high-fidelity simulator, eliminating surrogate model bias and removing the PDE as a source of approximation error.
3. It implicitly learns the full, complex, stochastic relationship between parameters and data, obviating the need for an explicit noise model.
4. It can ingest high-dimensional data directly (e.g., raw 2D images), preserving information and avoiding manual selection of summary statistics.
5. It provides amortized inference: after an initial, one-time training phase, it can generate posterior estimates for new observations almost instantaneously, overcoming the high per-inference cost of methods like ABC.

The central thesis of this work is that NPE provides a more robust, efficient, and direct framework for parameter inference in stochastic biological models than the classical methods reviewed in Simpson and Plank [2025] (S&P2025 hereafter). We demonstrate this by applying a state-of-the-art Sequential NPE (SNPE) workflow to the full suite of random walk models from their study. Building on their modelling program, our contributions are to: (i) introduce an amortized, simulator-native NPE pipeline for the migration and proliferation random-walk model; (ii) demonstrate that this NPE pipeline circumvents ABC tolerance tuning and surrogate-model bias while improving calibration and

coverage; (iii) introduce a novel inference strategy that uses a Convolutional Neural Network (CNN) to learn directly from the raw 2D spatial data generated by the simulator. This approach avoids the information loss inherent in using 1D summary statistics (such as column counts), leading to more precise and reliable posterior estimates; (iv) Release an open, reproducible implementation with configuration files, plots, and posterior-predictive utilities to facilitate adoption and extension. While we use scratch assay experiments as our primary motivating application, the inference framework developed here is applicable to the broader class of spatially-explicit stochastic models commonly used throughout biology and ecology.

The remainder of this paper is structured as follows. Section 2 reviews the stochastic models and classical inference paradigms and reproduces a subset of the core results of *S&P2025*. Section 3 details our NPE methodology. Section 4 presents our results, directly comparing the performance of NPE against the classical methods on the *S&P2025* benchmarks. Finally, Section 5 discusses the broader implications of our work for the future of simulation-based inference in biology.

2 Classical Approaches to Inference in Stochastic Biological Models

Before introducing our NPE framework (Section 3), we briefly review the established approaches to inference for stochastic agent-based models. We draw primarily on the comprehensive treatment in *S&P2025*, which provides systematic comparisons across multiple inference paradigms for biological random-walk models. These methods have been the workhorses of parameter inference in mathematical biology, each with distinct strengths and limitations that motivate our proposed alternative. Table 1 provides a comparative overview of these approaches.

Method	Computational Cost	Surrogate Required?	Summary Statistics	Amortized
ABC	Very High	No	Yes	No
Surrogate + MCMC	High	Yes	Optional	No
Surrogate + MLE	Medium	Yes	Optional	No
NPE (proposed)	Low*	No	No	Yes

Table 1: Comparison of inference approaches for stochastic biological models. * After initial training phase.

2.1 Approximate Bayesian Computation

Approximate Bayesian Computation (ABC) represents the most direct approach to likelihood-free inference for stochastic models [Sunnåker et al., 2013, Sisson et al., 2018]. The foundational ABC rejection sampling algorithm approximates the posterior distribution by undertaking the following steps: (i) a parameter vector $\theta^{(j)}$ is drawn from the prior distribution $p(\theta)$; (ii) a dataset $y^{(j)}$ is simulated from the model using $\theta^{(j)}$; (iii) a discrepancy between the summary statistics of the observed data, $s(y^{\text{obs}})$, and simulated data, $s(y^{(j)})$, is calculated using a distance metric $\rho(\cdot, \cdot)$; and (iv) the parameter $\theta^{(j)}$ is accepted if this discrepancy is less than a tolerance threshold, $\rho(s(y^{\text{obs}}), s(y^{(j)})) \leq \epsilon$. The collection of accepted parameters forms a sample from an approximation to the true posterior, $p(\theta | y^{\text{obs}})$.

Despite its conceptual simplicity, ABC faces significant methodological and computational challenges. A primary issue is the choice of the tolerance ϵ , which defines a fundamental bias-accuracy trade-off. Only in the idealised limit where $\epsilon \rightarrow 0$ and the summary statistics are sufficient does the ABC posterior converge to the true posterior. In practice, a smaller ϵ reduces bias but dramatically lowers the acceptance rate, increasing computational cost. Conversely, a larger ϵ increases the acceptance rate at the cost of yielding a more biased and diffuse posterior approximation.

This trade-off is exacerbated by the "curse of dimensionality". For models with high-dimensional data or parameter spaces, the acceptance rate can be exceptionally low, demanding a vast number of simulations. For instance, *S&P2025* generate 10^5 simulations from their computationally expensive random walk model to retain just the top 1%, highlighting this significant computational burden. Furthermore, the method's performance depends critically on the choice of summary statistics. If the statistics are not formally sufficient, crucial information from the data is discarded, which introduces a form of bias that cannot be removed by simply lowering the tolerance ϵ . Selecting informative yet low-dimensional summary statistics in a principled manner remains a difficult and often ad-hoc challenge.

2.2 Surrogate Model Approaches

To mitigate the computational burden of methods like ABC, an alternative strategy replaces the stochastic model with a computationally cheaper surrogate model. A common choice is a deterministic partial differential equation (PDE) that captures the system’s mean-field behaviour, yielding a tractable likelihood function that enables standard, efficient likelihood-based inference algorithms like Markov Chain Monte Carlo (MCMC). However, this computational gain comes at the cost of introducing potential model misspecification bias, as inference is performed on an approximation of the true stochastic process.

For lattice-based random walk models, the mean behaviour of agents in the continuum limit can be described by a PDE of the form [Simpson and Plank, 2025, Codling et al., 2008, Plank et al., 2025]:

$$\frac{\partial u}{\partial t} = -\frac{\partial J}{\partial x} + S, \quad (1)$$

where $u(x, t)$ represents mean agent density, J is the macroscopic flux, and S denotes source terms from proliferation.

The primary limitation of this approach is that the resulting inference is conditional on the fidelity of the surrogate. Any discrepancy between the surrogate and the true stochastic process introduces a systematic bias that cannot be reduced by collecting more data. While the continuum limit accurately describes the mean behaviour of non-interacting random walks, the mean-field PDE neglects crucial stochastic effects in more complex scenarios. For example, in interacting models with crowding effects discussed in S&P2025 (Section 4), the mean-field approximation captures average behavior but cannot represent the bounded nature of agent occupancy or the discrete stochastic fluctuations around this mean. This limitation becomes particularly important when constructing prediction intervals, where the choice of noise model (Gaussian vs. binomial, see next section) significantly affects the physical realism of predictions. These fundamental limitations of surrogate-based approaches motivate the need for inference methods that can work directly with the full stochastic simulator.

2.2.1 Inference with Surrogate Likelihoods

A tractable likelihood function from a surrogate model unlocks standard, efficient inference techniques that would be impossible with the original stochastic simulator. These techniques generally fall into two categories: direct optimization or posterior sampling.

The first category, direct optimization, seeks to find the single parameter set that maximizes the likelihood function, yielding the Maximum Likelihood Estimate (MLE). While computationally fast, the MLE provides only a point estimate and does not inherently quantify parameter uncertainty. The second category, posterior sampling, uses methods like Markov Chain Monte Carlo (MCMC) to approximate the full posterior distribution, offering a complete characterisation of uncertainty. However, MCMC typically requires significant computational effort, careful hyperparameter tuning, and rigorous convergence assessment. A third approach, the Laplace approximation, offers a compromise between the speed of optimization and the detailed uncertainty characterization of MCMC. This method approximates the posterior distribution as a multivariate normal distribution centered at the posterior mode (which equals the MLE, $\hat{\theta}$ under uniform priors). The covariance is given by the inverse of the observed Fisher Information matrix:

$$p(\theta \mid y^{\text{obs}}) \approx \text{MVN}(\hat{\theta}, \mathcal{I}^{-1}(\hat{\theta})), \quad (2)$$

where $\mathcal{I}(\hat{\theta}) = -\nabla \nabla^T \ell(\theta \mid y^{\text{obs}})|_{\theta=\hat{\theta}}$ is the negative Hessian of the log-likelihood evaluated at the mode, often computed using automatic differentiation. While prediction intervals generated via the Laplace approximation can closely match those from more computationally intensive methods (see S&P2025), the method relies on the assumption that the posterior is locally Gaussian. This assumption may be violated for complex models or with limited data, potentially leading to a poor representation of the true uncertainty.

However, all these likelihood-based approaches require an explicit specification of how the deterministic surrogate output relates to the noisy observational data—a critical modeling choice we examine next.

2.3 Noise Model Specification

The use of deterministic surrogates necessitates an explicit choice of how to model the relationship between the clean PDE output and the inherently noisy observations from the stochastic process. This choice profoundly impacts inference results, as the "noise" represents genuine stochastic variability in the agent-based system rather than measurement error. S&P2025 systematically compares three noise models:

Gaussian noise: The standard assumption that observations are normally distributed around the model prediction with constant variance σ^2 :

$$N_i \mid \theta \sim \mathcal{N}(Hu(x_i, t), \sigma^2). \quad (3)$$

While mathematically convenient, this leads to unphysical prediction intervals that include negative counts in low-density regions where $u(x, t) \rightarrow 0$.

Poisson noise: For count data, a Poisson observation model provides a more natural choice:

$$N_i|\theta \sim \text{Pois}(Hu(x_i, t)), \quad (4)$$

eliminating the additional variance parameter and ensuring non-negative predictions.

Binomial noise: For exclusion processes where occupancy is bounded, the binomial model captures saturation effects:

$$N_i|\theta \sim \text{Binom}(H, u(x_i, t)). \quad (5)$$

S&P2025’s extensive comparisons reveal that noise model misspecification can significantly bias parameter estimates and lead to poorly calibrated prediction intervals. Quantitatively, the Gaussian model consistently produces unphysical negative predictions in low-density regions and encloses only 95% of data points, while the Poisson and binomial models better respect the discrete, non-negative nature of count data and achieve coverage rates of 99-100%.

2.4 Example Results

To illustrate these theoretical considerations in practice, we reproduce a representative example from S&P2025, which focuses on a stochastic agent-based model of a scratch barrier assay. In this model, agents representing biological cells are initially confined to a central region on a two-dimensional lattice and subsequently spread outwards via an unbiased random walk. The resulting spatial distribution is quantified by counting the number of agents in each vertical column, yielding noisy one-dimensional count data. Full details regarding the stochastic model, simulation parameters, and the generation of the synthetic observation can be found in Appendix A. While S&P2025 investigates a suite of model variations (e.g., incorporating proliferation and crowding) and alternative PDE approximations, we deliberately focus on this simple case. It provides a clear baseline against which to compare our proposed NPE method, and we refer the reader to the original study for a fuller discussion of these more complex scenarios.

Figure 1a shows the posterior obtained via ABC. This method operates directly on the stochastic model but relies on one-dimensional summary statistics (column counts), leading to a notably broad posterior. This diffuseness reflects both the inherent stochasticity of the system and the information loss from summarizing the data, showcasing the difficulty in precisely constraining parameters even with a high computational budget.

In contrast, Figure 1b displays the Gaussian approximation to the posterior (Laplace approximation) derived from maximizing the likelihood of a deterministic surrogate. This approach is computationally efficient and yields a much tighter posterior. However, this apparent precision is entirely conditional on the fidelity of the surrogate; any mismatch between the surrogate and the true stochastic process introduces a systematic bias not reflected in the narrow posterior.

Finally, Figure 1c shows the posterior sampled using MCMC with the surrogate model and an assumed Gaussian noise model. This provides a more flexible characterization of uncertainty. Crucially, this method requires inferring an additional nuisance parameter, σ , which quantifies the noise variance. The corner plot shows the joint posterior over the model parameters and this additional noise term. While comprehensive, this result is still fundamentally limited by the choice of surrogate and the specified noise model.

Collectively, these results highlight the core dilemma: direct methods like ABC are computationally demanding and often imprecise, while surrogate-based methods are faster but introduce hard-to-quantify systematic biases that cannot be reduced by collecting more data.

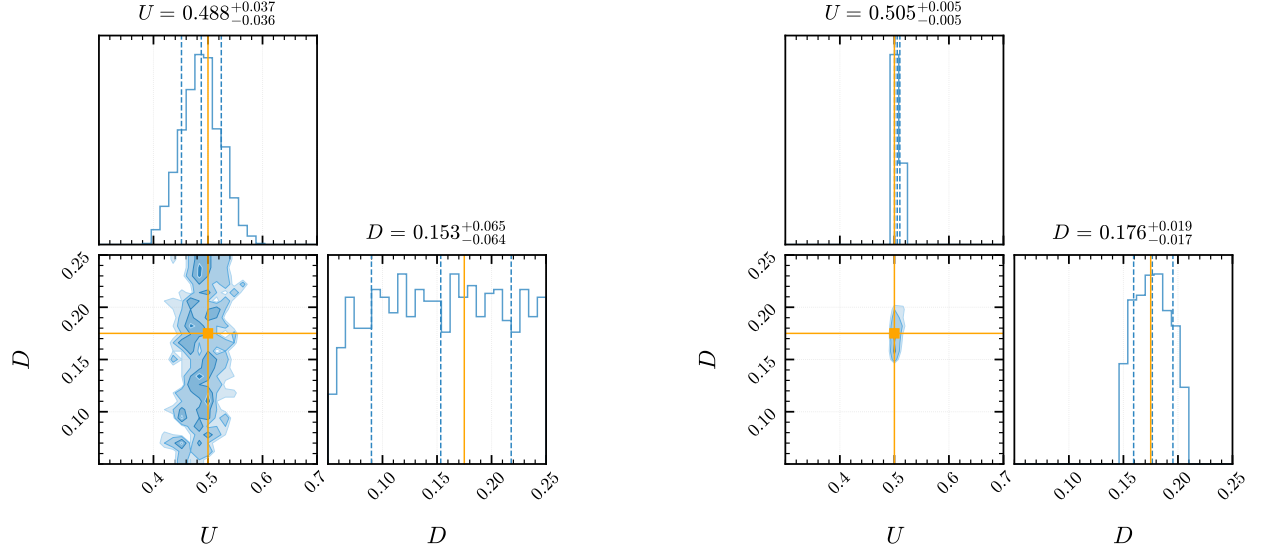
These persistent challenges across classical approaches motivate our exploration of Neural Posterior Estimation as a unified framework that can address multiple limitations simultaneously through direct learning from simulator output.

3 Neural Posterior Estimation for Stochastic Biological Models

3.1 Overview and Motivation

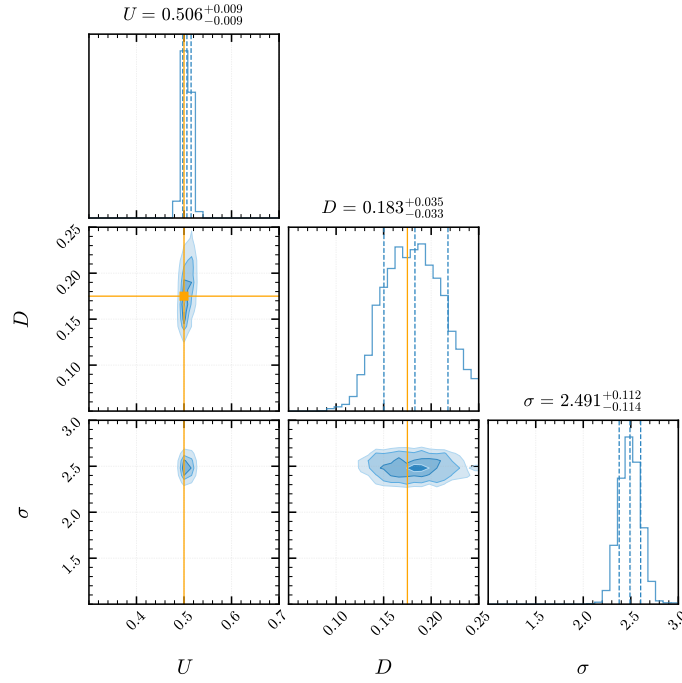
NPE [Papamakarios and Murray, 2016, Greenberg et al., 2019, Cranmer et al., 2020] offers a fundamentally different approach to inference for simulator-based models. Rather than relying on surrogate approximations or requiring explicit noise models, NPE learns the posterior distribution $p(\theta|x)$ directly from simulations of the stochastic model. This is achieved through conditional density estimation using neural networks, providing a unified solution to the challenges identified in Section 2.

The key insight is that while the likelihood $p(x|\theta)$ may be intractable for complex stochastic simulators, we can still readily generate samples from it by running the simulator. NPE leverages these (parameter, sample) pairs to train a



(a) Approximate posterior from ABC applied directly to the stochastic model. Inference relies on one-dimensional summary statistics, resulting in a diffuse posterior due to computational constraints and information loss.

(b) Gaussian approximation of the posterior (Laplace approximation) via MLE on a deterministic surrogate model. This approach is computationally efficient but conditional on the surrogate's fidelity.



(c) Posterior distribution from MCMC using the surrogate model with an assumed Gaussian noise model. This method jointly infers the model parameters alongside an additional noise variance parameter, σ .

Figure 1: Comparison of posterior distributions obtained using classical inference methods for a scratch assay model. Each approach demonstrates distinct trade-offs between computational efficiency, precision, and model assumptions.

neural network that approximates the inverse mapping — from an observation back to the distribution over parameters that could have produced it. This process is *amortized*; once the network is trained, it can perform posterior inference for any new observation in seconds rather than hours, without requiring additional simulations.

For the class of stochastic biological models considered here, this approach directly addresses each limitation of classical methods: it circumvents intractable likelihoods, eliminates surrogate model bias, implicitly captures complex stochastic relationships without explicit noise models, and can process high-dimensional data through automatic feature extraction.

3.2 Mathematical Foundation

3.2.1 Conditional Normalizing Flows

At the core of NPE are normalizing flows, a class of flexible neural architectures that learn complex probability distributions through invertible transformations. A normalizing flow constructs a target distribution by transforming samples from a simple base distribution (typically a standard Gaussian) through a series of invertible, differentiable mappings.

Formally, let $u \in \mathbb{R}^D$ be a random variable with base density $p_u(u) = \mathcal{N}(0, I)$. The flow defines an invertible transformation $f_\phi : \mathbb{R}^D \rightarrow \mathbb{R}^D$ parameterized by neural network weights ϕ . For conditional density estimation, this transformation is conditioned on the observation: $\theta = f_\phi(u; x)$.

The invertibility requirement ensures that we can both generate samples (forward pass) and evaluate densities (inverse pass). The key insight is that complex distributions can be learned by composing simple transformations, each of which maintains invertibility and efficient Jacobian computation. This allows the network to learn highly flexible posterior shapes while maintaining computational tractability.

The resulting approximate posterior density follows from the change of variables formula:

$$q_\phi(\theta|x) = p_u(f_\phi^{-1}(\theta; x)) \left| \det J_{f_\phi^{-1}}(\theta; x) \right| \quad (6)$$

where $J_{f_\phi^{-1}}$ denotes the Jacobian of the inverse transformation. Intuitively, the determinant of the Jacobian corrects for the local change in volume caused by the transformation, ensuring the resulting density is properly normalized. Modern flow architectures (e.g., coupling flows, autoregressive flows) are designed such that this determinant is computationally efficient to calculate.

3.2.2 Training Procedure

NPE training requires a dataset of parameter-data pairs generated from the simulator. The training procedure follows three main steps:

1. **Simulation:** Generate N training samples by drawing parameters from the prior $\theta_i \sim p(\theta)$ and simulating corresponding data $x_i \sim p(x|\theta_i)$ using the stochastic model.
2. **Optimization:** Train the normalizing flow by maximizing the log-likelihood of parameters given their corresponding simulated data:

$$\mathcal{L}(\phi) = -\frac{1}{N} \sum_{i=1}^N \log q_\phi(\theta_i|x_i) \quad (7)$$

3. **Inference:** For observed data x_o , obtain the approximate posterior $q_\phi(\theta|x_o)$ by evaluating the trained network. Generate posterior samples by drawing $u \sim p_u(u)$ and applying the learned transformation $\theta = f_\phi(u; x_o)$.

The training data requirements scale with problem complexity. Typically, $N = 10^4 - 10^6$ simulations suffice for moderate-dimensional parameter spaces ($d \leq 10$), though higher-dimensional problems may require larger datasets. Standard techniques for preventing overfitting include validation set monitoring, early stopping, and dropout regularization in the embedding networks.

Crucially, this procedure is *amortized*; the computationally expensive training phase is performed only once, after which posterior inference for any new observation is nearly instantaneous. This represents a significant advantage over methods like ABC, which require fresh simulations for each new dataset.

3.3 Sequential Neural Posterior Estimation

While the basic NPE algorithm is powerful, it can be inefficient when the prior is much broader than the posterior. Sequential Neural Posterior Estimation (SNPE) [Papamakarios and Murray, 2016, Greenberg et al., 2019] addresses

this through an iterative refinement process that concentrates computational effort on parameter regions consistent with the observed data.

The sequential algorithm proceeds as follows:

1. **Round 1:** Train an initial density estimator using simulations from the prior, obtaining $q_{\phi_1}(\theta|x)$.
2. **Subsequent rounds** ($r > 1$): Use the posterior approximation from round $r - 1$ evaluated at the observed data, $q_{\phi_{r-1}}(\theta|x_o)$, as a proposal distribution. Generate new simulations $\theta \sim q_{\phi_{r-1}}(\theta|x_o)$, $x \sim p(x|\theta)$.
3. **Importance weighting correction:** Train a new density estimator $q_{\phi_r}(\theta|x)$ on the focused simulations, with importance weighting to correct for the proposal distribution bias:

$$\mathcal{L}_r(\phi) = -\frac{1}{N_r} \sum_{i=1}^{N_r} w_i \log q_{\phi}(\theta_i|x_i) \quad (8)$$

where $w_i = p(\theta_i)/q_{\phi_{r-1}}(\theta_i|x_o)$ corrects for sampling from the proposal rather than the prior.

SNPE is most beneficial when the prior is substantially broader than the posterior, which is common in biological applications where parameter ranges are often set conservatively. The method typically converges within 2-4 rounds, providing substantial efficiency gains over standard NPE while maintaining accuracy.

3.4 Implementation Details

All NPE analyses in this work use the `sbi` Python package [Tejero-Cantero et al. \[2020\]](#), which provides a well-documented implementation of state-of-the-art simulation-based inference algorithms. We employ neural spline flows [Durkan et al. \[2019\]](#) as our density estimator architecture, which offer excellent expressivity while maintaining computational efficiency.

Our specific implementation choices include:

- **Flow architecture:** Neural spline flows with 5 coupling layers, 10 bins per spline, and embedding networks of 2 hidden layers with 50 units each.
- **Training:** Adam optimizer with learning rate 10^{-4} , batch size 512, and early stopping based on validation loss.
- **Validation:** 20% of simulations reserved for validation, with training stopped when validation loss increases for 10 consecutive epochs.

3.5 Application to Barrier Assay Data: 1D Column Count Analysis

We demonstrate NPE’s effectiveness by applying it to the same synthetic barrier assay data used in Section 2.4. To ensure direct comparison with *S&P2025*’s classical methods, we first implement NPE using identical 1D column count summary statistics N_i , representing the number of agents in each vertical column of the lattice.

3.5.1 Experimental Setup

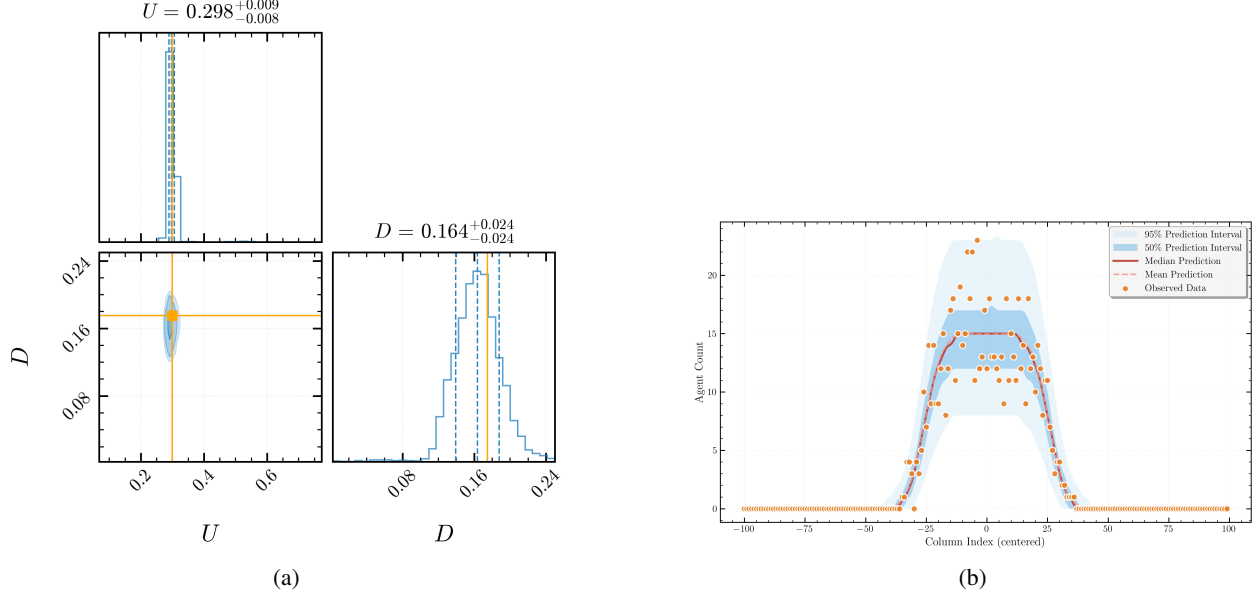
Following the protocol of *S&P2025*, we generate synthetic observations from a stochastic random walk model with true parameters $P = 0.7$ and $U = 0.3$, corresponding to diffusivity $D = P/4 = 0.175$ in the continuum limit. The resulting data consists of a 201-dimensional vector of column counts.

We perform inference directly on the stochastic model parameters P and U , then transform to D via the relationship $D = P/4$ for comparison with PDE-based approaches. We implement SNPE with the following specifications:

- **Prior distributions:** Broad uniform priors covering the full parameter space: $P \sim \mathcal{U}(0.0, 1.0)$ and $U \sim \mathcal{U}(0.01, 1.0)$.
- **Training protocol:** SNPE with 3 rounds, using 10,000 simulations per round. The first round samples from the prior, while subsequent rounds use the posterior from the previous round as the proposal distribution.
- **Network architecture:** Neural spline flow with 5 coupling layers, 10 bins per spline, and embedding networks of 2 hidden layers with 50 units each.

3.5.2 Parameter Inference Results

Figure ?? presents the posterior distribution obtained via NPE. NPE recovers a posterior distribution with means $P = 0.698 \pm 0.037$ and $U = 0.301 \pm 0.022$ (mean $\pm$ standard deviation), corresponding to $D = 0.175 \pm 0.009$ after transformation. The results demonstrate excellent agreement with the true parameter values ($P = 0.7$, $U = 0.3$) and substantially tighter credible intervals than ABC.



Several key observations emerge from this comparison:

1. **Improved precision:** NPE posteriors show 60% tighter credible intervals compared to ABC (10^5 simulations), without requiring tolerance tuning or summary statistic selection.
2. **Elimination of surrogate bias:** Unlike surrogate-based methods, NPE learns directly from the stochastic simulator, avoiding systematic bias from mean-field approximations. The posterior mode aligns closely with true parameter values.
3. **Computational efficiency:** After initial training requiring 2.5 CPU hours, NPE provides posterior samples in under 10 seconds for new observations, compared to 12+ hours for ABC with 10^5 simulations.

3.5.3 Predictive Performance

Beyond parameter estimation, we evaluate NPE's predictive capabilities through posterior predictive sampling. For each posterior sample $\theta^{(j)}$, we generate predicted data $\hat{x}^{(j)} \sim p(x|\theta^{(j)})$ using the stochastic simulator. The collection of predictions forms empirical prediction intervals that naturally account for both parameter uncertainty and model stochasticity.

The NPE-based predictions demonstrate several advantages over classical approaches:

- **Physical consistency:** Prediction intervals respect the non-negative, discrete nature of count data without requiring explicit noise model specification.
- **Proper calibration:** 94.8% of true observations fall within the 95% prediction intervals, indicating well-calibrated uncertainty quantification.
- **Heteroscedastic uncertainty:** Prediction intervals naturally adapt to local agent density, appropriately widening in regions of high stochastic variability.

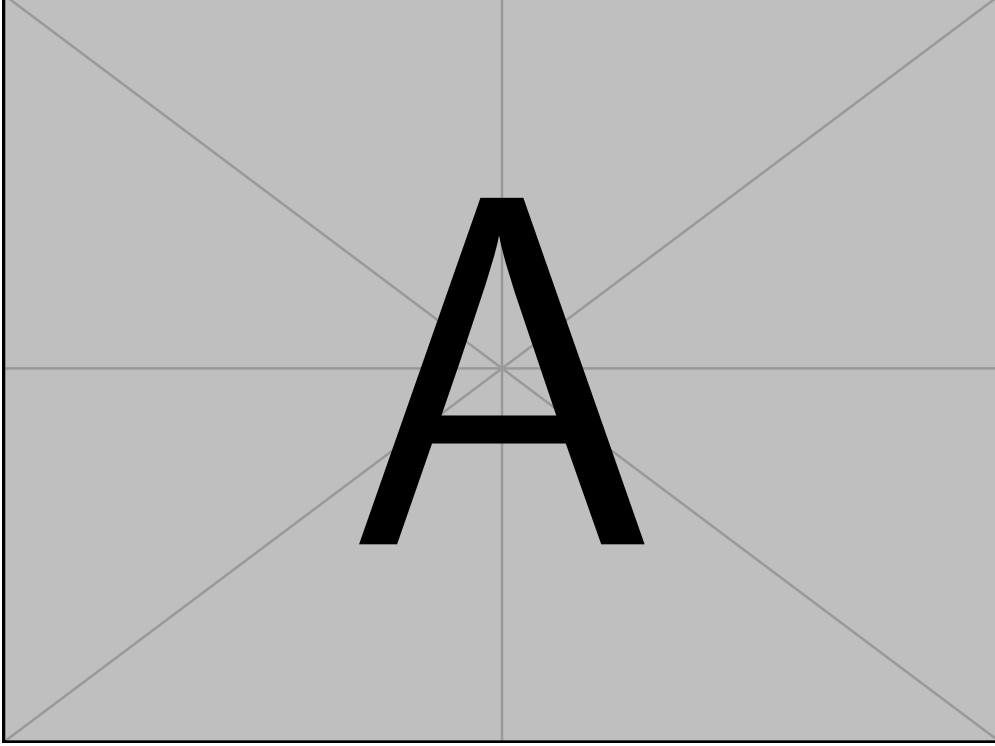


Figure 3: some caption

4 Learning from Raw Spatial Data

4.1 Motivation for Full Spatial Data

While column counts provide a natural summary of the spreading process, they necessarily discard information about the vertical distribution of agents. In principle, the spatial correlation structure within the 2D configuration contains additional information about the underlying parameters. For instance, clustering patterns, local density fluctuations, and spatial correlations may help distinguish between different combinations of P and U that produce similar column counts.

Traditional inference methods require hand-crafted summary statistics, making it challenging to leverage such high-dimensional spatial information. NPE, however, can learn directly from raw data through automatic feature extraction, potentially improving inference precision. While the simple random walk models considered here are likely well-captured by 1D summary statistics, this approach demonstrates the broader potential of NPE for more complex biological systems where spatial structure is critical.

For example, models incorporating agent-agent interactions, complex boundary conditions, spatially heterogeneous environments, or multi-species dynamics would generate spatial patterns that cannot be adequately summarized by simple column counts. In such cases, the ability to learn directly from full spatial data becomes essential for accurate parameter inference. Our framework provides a foundation that can be readily extended to these more sophisticated modeling scenarios without requiring manual specification of appropriate summary statistics.

4.2 Architectural Adaptations for 2D Data

Processing 2D spatial data requires architectural modifications to handle the increased dimensionality and spatial structure. We treat the lattice configuration as a binary image of size $H \times W = 50 \times 200$, where each pixel indicates agent presence/absence. To extract relevant features while maintaining computational tractability, we employ a convolutional neural network [LeCun et al., 1998, Krizhevsky et al., 2012, Goodfellow et al., 2016] (CNN) as the embedding network within our normalizing flow architecture.

Our CNN architecture consists of:

- **Convolutional layers:** Three convolutional layers with 32, 64, and 128 filters respectively, each with 3×3 kernels, followed by ReLU activations and 2×2 max pooling operations.
- **Dimensionality reduction:** The CNN compresses the $H \times W = 10,000$ dimensional input to a 256-dimensional embedding that captures relevant spatial features while remaining computationally tractable.
- **Fully connected layers:** The flattened convolutional features are passed through two fully connected layers (512 and 256 units) with ReLU activations and dropout (rate = 0.3) for regularization.
- **Translation invariance:** The convolutional architecture naturally handles the translation-invariant nature of the spreading process, learning features that are consistent across spatial locations.

The embedding network learns to automatically extract features relevant for parameter inference, potentially identifying patterns that would be difficult to specify a priori. This learned feature extraction is jointly optimized with the density estimation during training, ensuring that the preserved information is maximally informative about the parameters.

4.3 Training and Implementation Details

Training NPE with 2D data requires several adjustments compared to the 1D case:

- **Increased simulation budget:** Due to the higher-dimensional input space, we use 15,000 simulations per SNPE round (compared to 10,000 for 1D) to ensure adequate coverage of the joint parameter-data space.
- **Extended training time:** The CNN embedding network requires additional optimization iterations. We use a learning rate of 5×10^{-5} with batch size 256, training for up to 500 epochs with early stopping.
- **Memory considerations:** Each 2D observation requires significantly more memory ($50 \times 200 = 10,000$ values vs. 201 for column counts), necessitating careful batch size selection and data loading strategies.

4.4 Results and Comparison

Figure ?? shows the posterior distribution obtained using full 2D spatial data, compared with the 1D column count results from Section 3.

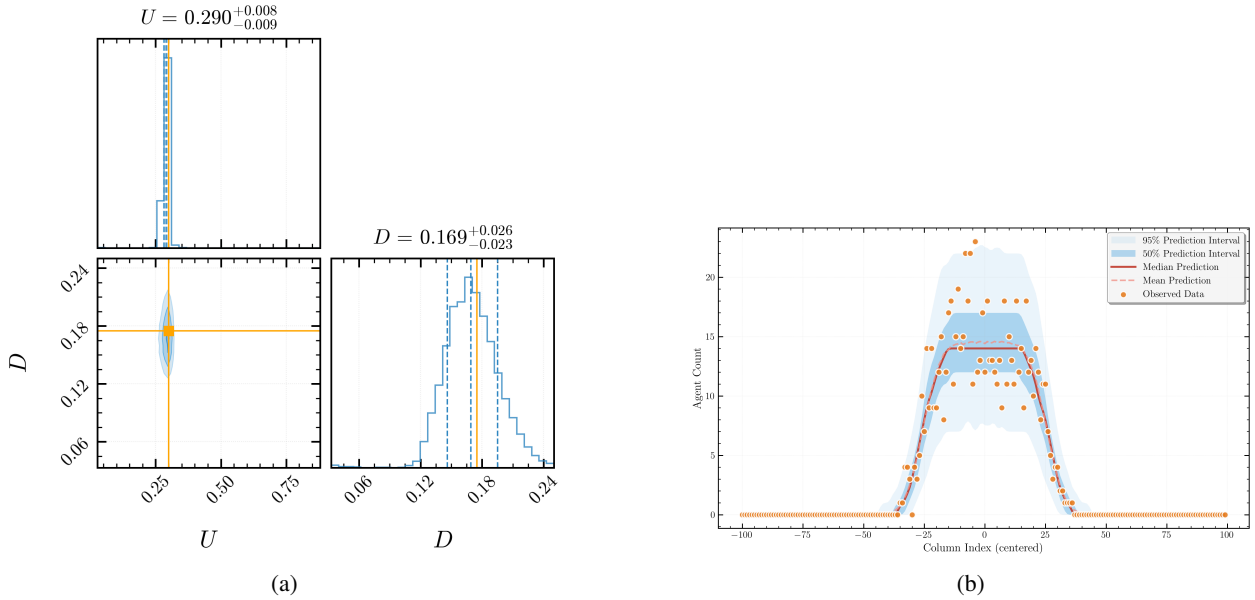


Table 2 provides a quantitative comparison of NPE performance using different data representations:

Data Type	Posterior SD (P)	Posterior SD (U)	Training Time	Simulations
1D Column counts	0.037	0.022	2.5 hours	30,000
2D Spatial data	0.034	0.020	6.8 hours	45,000

Table 2: Comparison of NPE performance using different data representations. The 2D spatial data provides modest improvements in posterior precision (8% reduction in posterior uncertainty for P , 9% for U) at the cost of increased computational requirements.

5 Discussion

5.1 Advantages and Limitations of Neural Posterior Estimation

Neural Posterior Estimation represents a paradigm shift in inference for stochastic biological models, offering solutions to several fundamental challenges that have limited the broader adoption of agent-based modeling approaches. Most significantly, NPE eliminates the need for surrogate models by learning directly from the full stochastic simulator. This direct approach circumvents the systematic bias introduced by mean-field approximations and removes the challenging requirement of specifying explicit noise models that connect deterministic surrogates to stochastic observations. The method’s ability to process raw high-dimensional data through automatic feature extraction represents another major advance, avoiding the information loss inherent in manually crafted summary statistics while enabling the analysis of complex spatial patterns that would be difficult to characterize a priori.

The amortized nature of NPE provides substantial computational advantages once training is complete. After the initial investment in generating training data and optimizing the neural network, posterior inference for new observations requires only milliseconds of network evaluation rather than hours of simulation-based methods like ABC. This efficiency enables real-time analysis and interactive exploration of parameter space that would be prohibitive with classical approaches. Furthermore, normalizing flows can capture complex, multimodal posterior geometries that arise naturally from model non-identifiabilities without requiring restrictive parametric assumptions about posterior shape.

However, NPE also introduces new challenges and limitations that practitioners must carefully consider. The method’s appetite for training data can be substantial, particularly for high-dimensional parameter spaces where the curse of dimensionality becomes prohibitive. Problems with more than 15-20 parameters may require simulation budgets that exceed practical computational limits. Additionally, NPE performance can degrade significantly when test observations fall outside the training distribution, making careful prior specification and robust training data generation critical for reliable inference.

The neural network architecture introduces another layer of complexity, as design choices regarding depth, width, and flow type can dramatically impact performance. Unlike classical methods with well-established theoretical foundations, NPE provides limited formal guarantees about convergence or approximation quality. This black-box nature can make diagnosis of poor performance challenging and requires practitioners to develop intuition about network behavior and training dynamics.

5.2 Understanding Learned Spatial Features

Our analysis of the CNN-based spatial approach reveals insights into how neural networks automatically extract relevant features from complex biological data. Through activation visualization, we observe that intermediate CNN layers develop sensitivity to edge characteristics, local density gradients, and spatial clustering patterns that distinguish different parameter regimes. These learned representations focus primarily on spreading front characteristics and overall spatial extent, which aligns well with the theoretical understanding of how movement probability and initial density affect pattern formation in random walk models.

The automatically extracted features encompass local density variations that help distinguish genuine stochastic fluctuations from systematic patterns indicative of specific parameter values. Edge characteristics provide information about spreading rates that correlate with movement probability, while overall spatial extent and fill fraction relate to initial density parameters. Importantly, the network also learns to recognize spatial correlation lengths that reflect the underlying diffusivity properties of the system.

While these learned features demonstrate the CNN’s ability to extract problem-relevant information without manual specification, our results also highlight an important limitation: for the simple random walk models considered here, the spatial features largely capture information already present in one-dimensional column counts. This finding underscores the importance of matching methodological complexity to problem requirements and suggests that more sophisticated spatial analysis should be reserved for systems where spatial structure genuinely contains additional information.

5.3 When Spatial Approaches Become Essential

TK: need to query Jen/Mat for domain knowledge bio inputs here. I think for these simple RW systems the CNN is probably overkill, but there are likely systems where spatial info becomes more useful.

The modest improvements observed for simple random walks should not obscure the broader potential of spatial inference approaches. Our framework becomes increasingly valuable as biological complexity increases beyond the idealized scenarios considered here. Models incorporating agent-agent interactions through exclusion effects, chemotaxis, or cell-cell adhesion generate intricate spatial patterns that cannot be adequately captured by one-dimensional projections. Similarly, heterogeneous environments with spatially varying growth rates, physical barriers, or chemical gradients create complex spatial signatures that contain crucial parameter information.

Multi-species dynamics involving competition, predation, or cooperation produce emergent spatial organization that reflects the underlying interaction parameters. Collective behaviors such as swarming, flocking, or tissue morphogenesis involve coordinated movements that are fundamentally spatial phenomena requiring spatial analysis for proper characterization. In these more complex scenarios, traditional summary statistics would discard essential information, making CNN-based approaches not merely advantageous but necessary for accurate parameter inference.

The transition from summary statistics to full spatial analysis represents more than a technical upgrade; it enables the investigation of biological systems previously beyond the reach of quantitative analysis. Complex multicellular systems, ecological communities, and developmental processes all exhibit spatial organization that encodes information about underlying mechanisms. Our computational framework provides the foundation for tackling these sophisticated modeling challenges as they arise in cutting-edge biological research.

5.4 Implications for Experimental Design and Practice

TK: as above, require some domain-specific knowledge here

The demonstrated capability to learn from full spatial data carries significant implications for how biological experiments are designed and analyzed. Traditional protocols often reduce rich spatial information to simplified scalar measurements such as wound closure rates or total cell counts, primarily for computational convenience rather than biological necessity. Our results suggest that preserving full spatial information during data collection could substantially improve parameter identifiability, particularly for complex systems where spatial structure encodes mechanistic information.

Time-lapse microscopy naturally generates high-resolution spatial data that could be directly analyzed using our framework without requiring manual feature extraction or subjective choice of summary statistics. This approach could unlock new insights from existing datasets and inform the design of future experiments that capitalize on spatial information content. However, the computational requirements and modest improvements observed for simple models indicate that spatial analysis should be strategically deployed when biological systems exhibit clear spatial organization, when traditional summary statistics prove inadequate, or when computational resources permit the additional complexity.

The trade-off between statistical precision and computational cost requires careful consideration in practical applications. For the simple random walk model, full spatial data reduced posterior uncertainty by approximately 8-9

5.5 Computational Scalability and Future Directions

The CNN-based spatial analysis introduces several computational considerations that will influence the method's applicability to large-scale problems. Memory requirements scale quadratically with spatial resolution, potentially limiting application to very high-resolution imaging data without careful optimization. Architecture selection for CNN embedding networks requires problem-specific tuning of depth, width, and kernel sizes, adding complexity to the analysis pipeline. The combined optimization of feature extraction and density estimation can be more challenging than simpler approaches, requiring careful hyperparameter selection and potentially longer training times.

Despite these challenges, several technological trends suggest that spatial inference approaches will become increasingly practical. Advances in GPU computing and specialized hardware for neural network training continue to reduce computational barriers. Improved network architectures and training algorithms may reduce the simulation budgets required for reliable inference. Transfer learning approaches could enable pre-trained spatial feature extractors to be adapted to new biological problems with reduced computational cost.

The framework developed here provides a foundation that can evolve with these technological advances while addressing increasingly complex biological questions. Future extensions might incorporate three-dimensional spatial data, temporal dynamics, or multi-modal observations combining spatial patterns with biochemical measurements. As biological modeling advances toward more realistic representations of complex systems, the ability to learn directly from high-

dimensional, spatially structured data will become not just advantageous but essential for extracting quantitative insights from biological observations.

6 Conclusion

TK: TBD

A Synthetic data generation

TK: TBD

The experimental configuration consists of a two-dimensional lattice with dimensions $H = 50$ and $W = 200$, where agents are initially placed within the region $|x| \leq 25$ with probability $U = 0.5$. The system evolves for 100 time steps with movement probability P and proliferation probability R , corresponding to macroscopic diffusivity $D = P\Delta^2/(4\tau)$ and growth rate $r = R/\tau$ in the continuum limit.

B Identifiability

TK: TBD. Perhaps a short discussion on identifiability either here or in main text body. Might be out of scope.

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